



STAT 5228 - Data Exploration and Visualisation

Final Coursework Portfolio (Group B)

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Task 1 : Data audit, tidy data and cleaning

For Task 1, we have loaded the csv file in R software and has conducted the data audit, data tidying and cleaning. It is saved as cleaned_data.csv . We then filtered by district and product category and saved the focus data as Moka_data.csv.

The following table shows the extract of the variables and the issues we came across. (Full table: Table 1: Cleaning Log)

SN	Variable	Issue
1	Transaction_id	Duplication of ids
2	discount_pct	Missing values Values outside discount range (0-0.35)
3	quantity, unit price MUR	Included invalid values
4	gross_sales_mur, net_sales_mur, cost_mur, gross_profit_mur	Financial subtle inconsistencies
5	delivery times	Missing for in store
6	net sales mur	Outliers present
7	Date/datetime	Inaccuracy data format-data structure as character
8	satisfaction score	Values missingness
9	store_lat, store_lon	Store coordination missing

The initial dataset had 24000 observations; after applying all the cleaning measures we were left with 23511 observations.

Focus Brief

Our Focus District is **Moka** which is an operational district with an overall number of transactions of 3755 and has the minimum requirement of 2 stores. Our Focus category is **Electronics** which we have initially obtained by the calculation of the group leader's Student ID. We have decided to stick with Electronics because in comparison with other product categories, it is associated with the higher end of value transactions therefore we can have a more impactful sales and profit analysis.

Therefore, our filtered dataset has 423 observations. While checking for any missingness in our focus data we have come across the following: (full table: Table 2: Moka Missingness values)

delivery_minutes	satisfaction_score
258	45

From the above table, we can see that for delivery minutes and satisfaction score we have 258 and 45 observations respectively. This is not a mistake, instead it shows the structural missingness for the district of MOKA for in-store transactions. As for the satisfaction score of non-responsiveness being only 45, we have kept it unchanged to avoid biasness.

Task 2 : Univariate EDA + visual design

2.1 Overview

This univariate exploratory data analysis (EDA) examines the statistical characteristics of the dataset focusing on the Moka district and the Electronics product category. The purpose of this analysis is to understand the distribution, variability and statistical behaviour of the variables before conducting further modelling.

The dataset contains 423 observations. The main numerical variables analysed include Net Sales, Gross Profit, Delivery Minutes and Satisfaction Score. The categorical variables include Payment Method, Customer Gender, Customer Segment and Channel.

2.2 Numeric Distributions

This section examines the distributional characteristics of the main numeric variables using histograms, density plots, skewness, and kurtosis. The objective is to assess normality, detect skewness, and determine whether transformations are necessary for modelling.

2.2.1 Net Sales (MUR)

The distribution of Net Sales is approximately symmetric with a slight concentration around the central values. The histogram shows no extreme distortion, and the density curve confirms a fairly balanced shape. Skewness is close to zero, indicating minimal asymmetry in the data. This suggests that sales values are relatively stable across transactions, with no strong dominance of extreme high or low values.

Overall, Net Sales can be considered well-behaved for exploratory analysis, with no need for transformation at this stage.

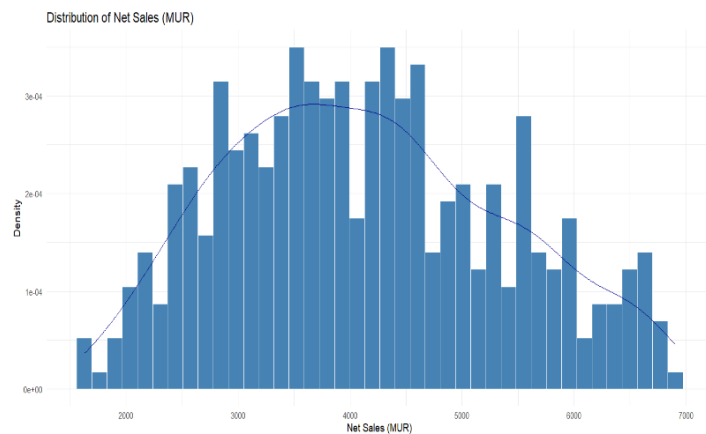


Figure 1: Net Sales Distribution

2.2.2 Gross Profit (MUR)

Gross Profit shows moderate positive skewness, with most values clustered in the mid-range and a smooth right tail reflecting higher-value transactions. The distribution is unimodal and well-structured, with no significant outliers. Overall, it remains suitable for interpretation in its original scale for univariate analysis.

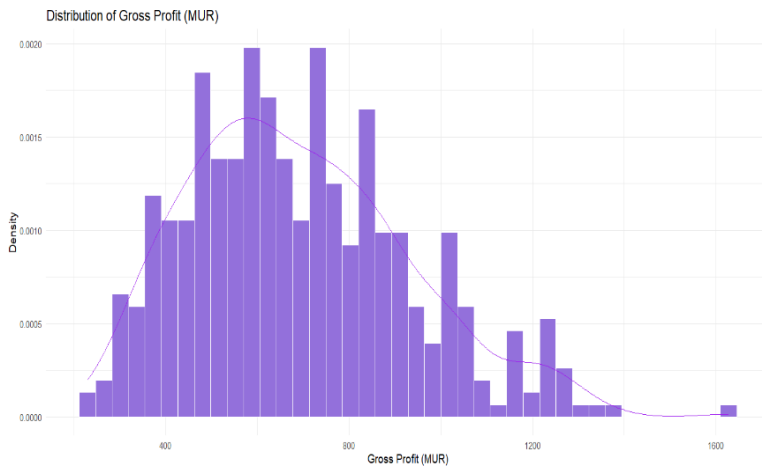


Figure 2: Gross Profit Distribution

2.2.3 Delivery Minutes

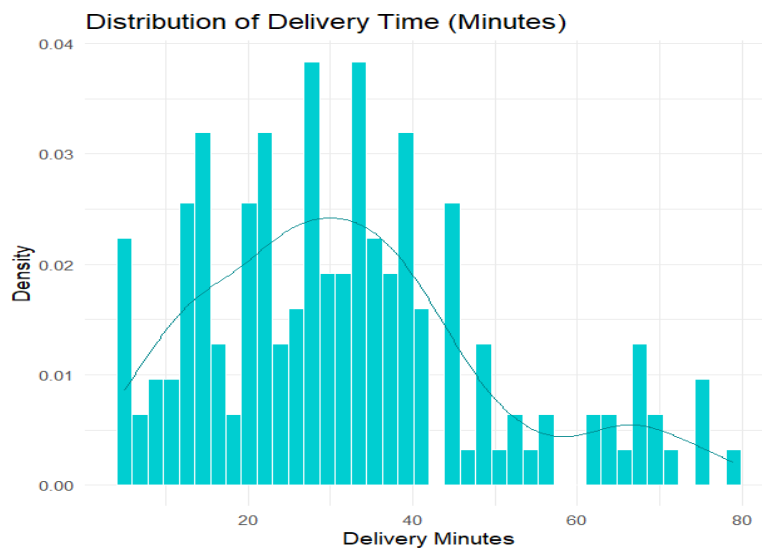


Figure 3: Delivery Minutes Distribution

The distribution of delivery time is moderately positively skewed, with most values clustered between 20 and 40 minutes and a gradual right tail toward longer durations. However, the skewness is not severe, and the distribution remains smooth and unimodal without extreme outliers. Therefore, a log transformation is not necessary, as the data is sufficiently well-behaved for analysis in its current form.

2.2.4 Satisfaction Score

Satisfaction Score is nearly perfectly symmetric with no extreme values. The skewness and kurtosis further confirm the absence of extreme response patterns, so no transformation is required.

Summary Statistics Table (Refer to

Task 2: Summary Statistics Table

Table 3) for Skewness and Kurtosis values

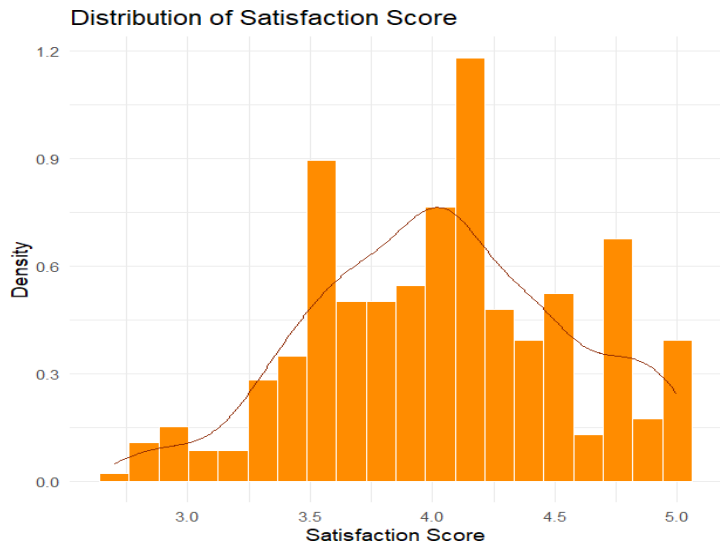


Figure 4: Satisfaction Score Distribution

2.3 Outlier Analysis (Boxplots)

Outliers in the boxplots are identified as individual points beyond the whiskers, which represent values more than 1.5 times the interquartile range (IQR) from the box. In this case, `delivery_minutes` shows a high outlier above the main distribution, while `gross_profit_mur` also displays an outlier above its upper whisker. In contrast, `net_sales_mur` shows no distinct outlier points, indicating that although it has a wide spread, its values remain within the expected range.

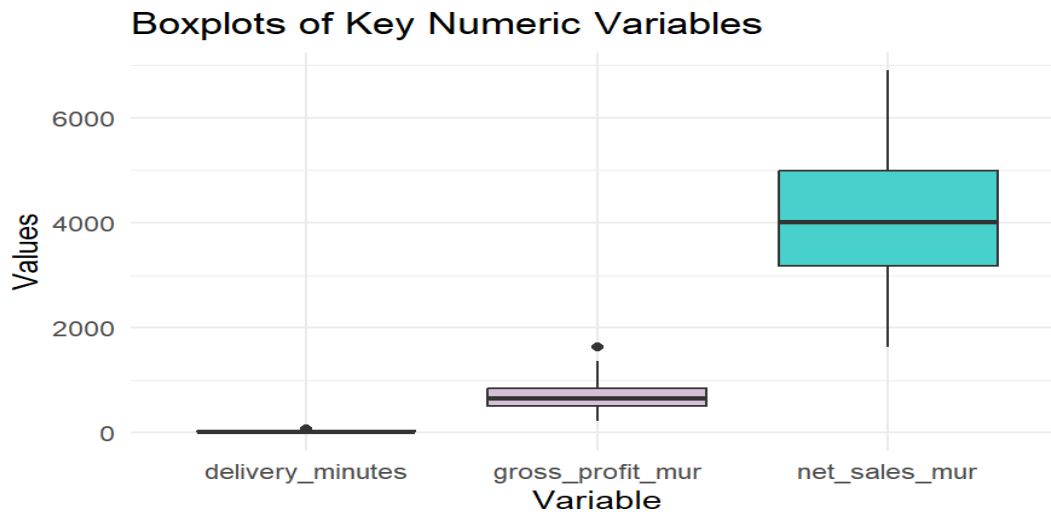


Figure 5: Boxplots for outlier

2.4 Categorical Distributions (Bar Charts)

The distribution of key categorical variables are analyzed using bar charts to identify dominant patterns and underlying customer behaviour across different dimensions. The following are concluded :

1. Card payments dominate, indicating strong preference for convenient payment options.

- Gender distribution is relatively balanced, with a slight predominance of female customers. This suggests no major gender disparity in purchasing behaviour
- Returning customers form the largest segment, followed by new customers, with VIP customers being the smallest group. This indicates strong customer retention and a stable revenue base.

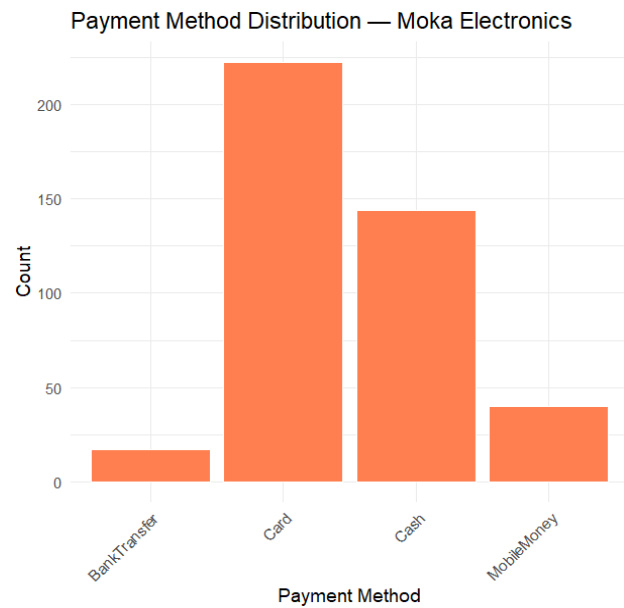


Figure c:Payment Method

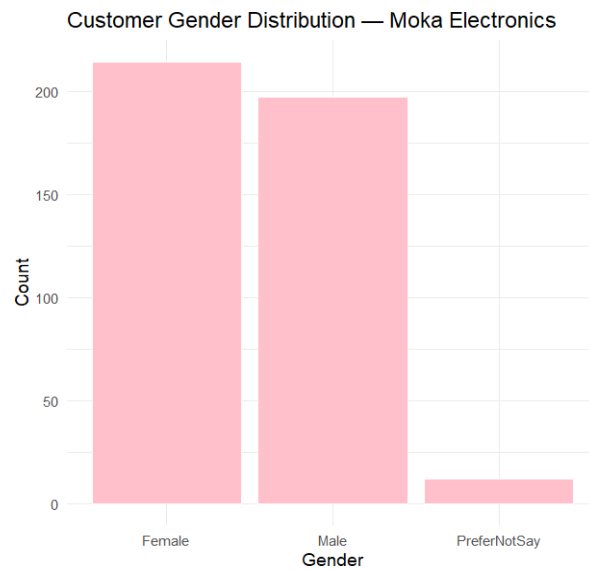


Figure 7: Customer as per Gender

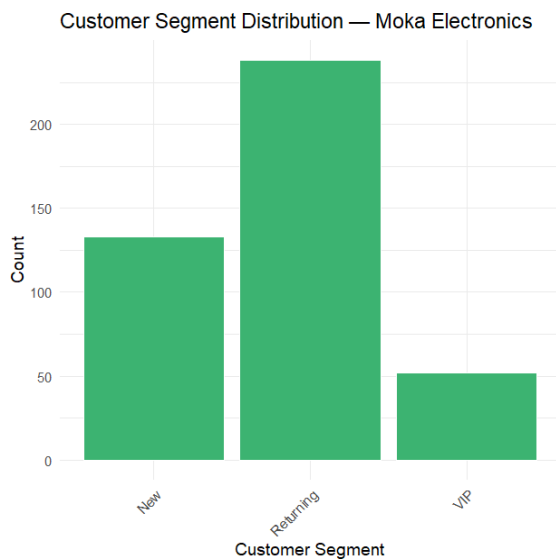


Figure 8: Group Segment

Conclusion

The univariate analysis shows that the dataset is generally well-structured and suitable for further analysis. The numerical variables are mostly well-behaved, with Net Sales and Satisfaction Score showing approximately symmetric distributions, while Gross Profit and Delivery Minutes display mild positive skewness due to higher transaction values and longer delivery times. A small number of outliers are present, mainly in Gross Profit and Delivery Minutes, reflecting normal operational variation. The categorical analysis shows a strong preference

for card payments, a dominant returning customer base, and a relatively balanced gender distribution. Differences in Delivery Minutes across channels highlight variation in operational efficiency. Overall, the dataset is consistent, reliable, and appropriate for further analysis.

Task 3 : Bivariate + multivariate EDA

3.1 Overview

This section presents the bivariate and multivariate exploratory data analysis (EDA). The aim is to understand how different variables are related and how they affect business performance and customer behavior.

The analysis focuses on key variables such as net sales, gross profit, and customer satisfaction. It also compares results across stores, sales channels, and customer characteristics like gender and age group.

3.2 Net Sales by Store and Channel

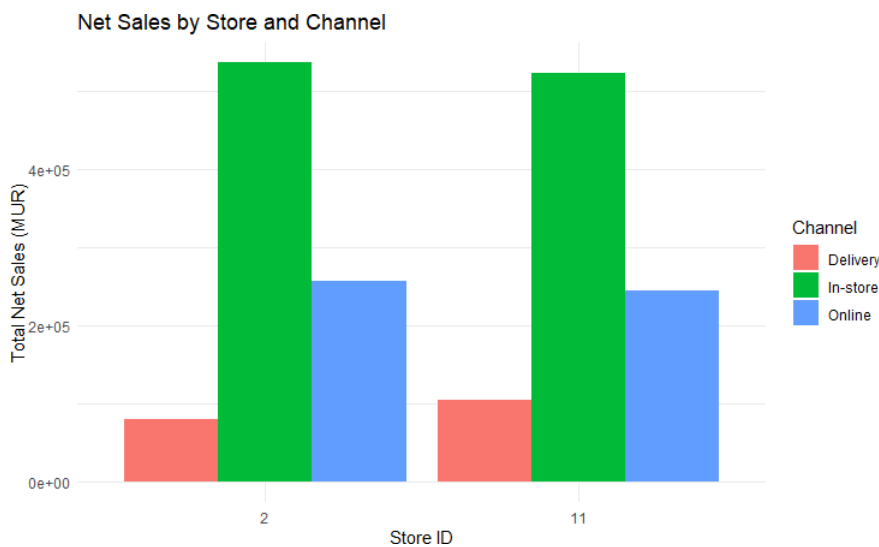


Figure 5: Net Sales by Store and Channel

The bar chart reveals a significant reliance on physical retail within the IslandMart ecosystem, as in-store sales dramatically outperform both online and delivery channels at Store 2 and Store 11. Specifically, in-store revenue for both locations comfortably exceeds the MUR 400,000 threshold, suggesting that the tactile shopping experience remains the primary driver of customer engagement in these districts. While Store 2 shows a slight lead in total volume and online traction, Store 11 demonstrates a marginally stronger performance in the delivery sector, potentially indicating a more effective local logistics set up or a more convenience-oriented demographic. However, the substantial gap between physical and digital sales highlights a clear opportunity for digital transformation and growth. To make this visual management-ready, the current scientific notation on the Y-axis should be converted to simplified currency units like "400k," and the store IDs should be replaced with district names to provide immediate geographical context for strategic decision-making.

3.3 Net Sales by Gender and Store

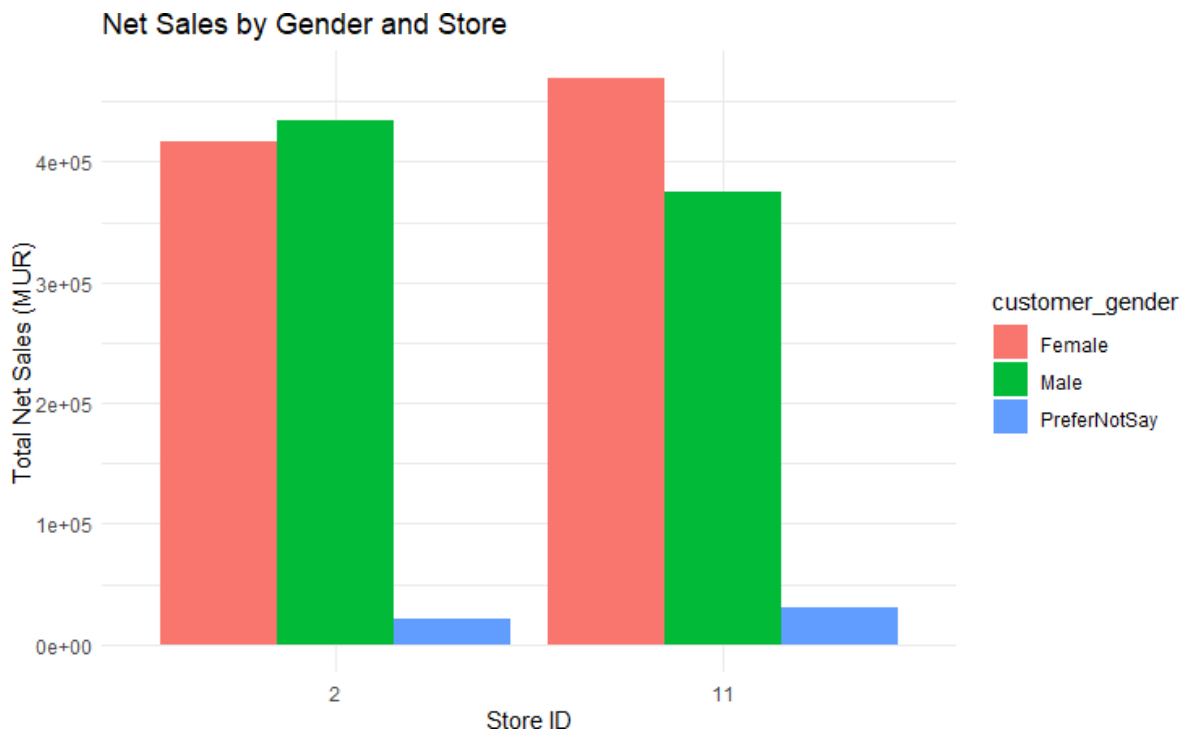


Figure 10: Net Sales by Gender and Store

The bar chart provided offers a detailed demographic breakdown of net sales across IslandMart locations, revealing distinct customer purchasing behaviors between Store 2 and Store 11. At Store 2, there is a relatively balanced distribution of revenue between male and female customers, with males showing a slight lead in total spending. In contrast, Store 11 exhibits a much more pronounced gender disparity, where female customers represent the dominant revenue stream, significantly outperforming their male counterparts. This shift suggests that the product mix or local marketing strategies in Store 11 are resonating more effectively with a female audience. Furthermore, both stores record a minimal but consistent contribution from customers who prefer not to disclose their gender, indicating a segment of the market that values privacy but currently represents a smaller fraction of the total MUR. From an analytical perspective, this visualization highlights the necessity of tailoring store-level inventory and promotional campaigns to match the specific demographic profiles of each district to maximize profitability.

Additional Plots such as violin plots are included in [Appendix 4](#).

3.4 Multivariate Relationships between key variables

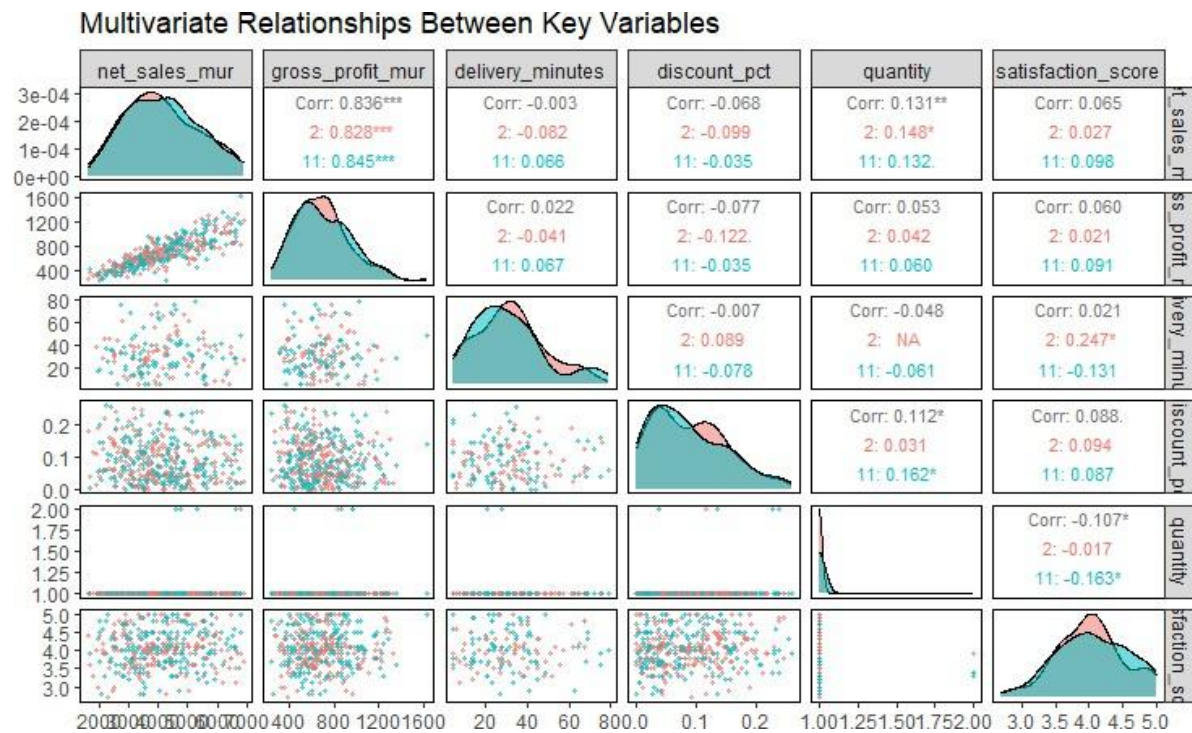


Figure 11: Multivariate Correlation Matrix

This correlation matrix provides a sophisticated multivariate analysis of IslandMart’s operational health, specifically isolating the drivers of profitability and customer experience for Store 2 and Store 11. The most striking insight is the exceptionally strong positive correlation of 0.836 between net sales and gross profit, which remains consistent across both locations. This linear relationship confirms that top-line growth is effectively translating into bottom-line returns without significant margin erosion. Interestingly, delivery speed appears to have a negligible impact on overall sales volume, suggesting that customers may be less sensitive to delivery minutes than expected, or that logistics are currently operating within an acceptable baseline. Furthermore, the satisfaction scores show a surprisingly weak correlation with discount percentages, implying that IslandMart customers might value product availability or service quality over aggressive price-cutting. These findings are vital for management as they suggest that increasing profitability should focus on volume and high-margin product categories rather than relying solely on delivery speed or heavy discounting to drive customer sentiment.

3.5 Product Subcategory by Age Band per Store

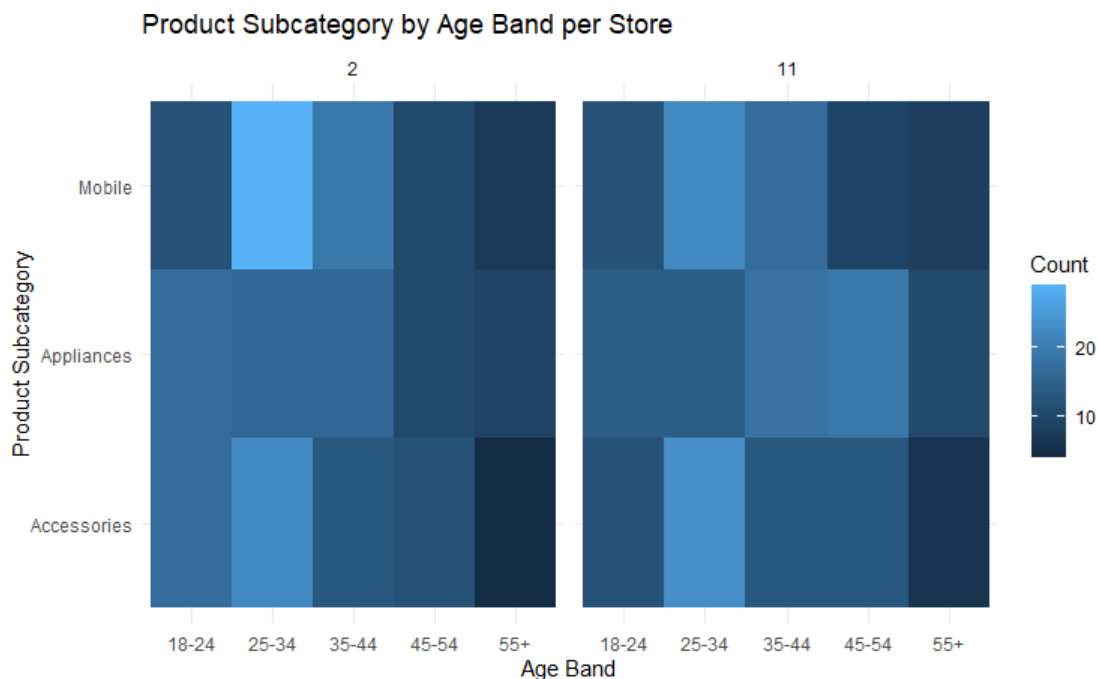


Figure 12: Heatmap

This heat map provides a multivariate view of IslandMart's customer demographics, crossing product subcategories with age bands across Store 2 and Store 11. A striking trend emerges in the "Mobile" category, where the 25-34 age demographic represents the most intense cluster of activity, particularly in Store 2. This suggests that younger professionals are the primary drivers of tech-related revenue for the chain. In contrast, the "Appliances" and "Accessories" categories show a more diffused engagement across middle-age brackets, specifically the 35-54 groups, indicating that household-oriented purchases are more evenly distributed as customers mature. Store 11 appears to have a slightly more balanced distribution across age bands compared to the sharper peaks seen in Store 2, which could imply a more diverse neighborhood footfall. For management, these insights are instrumental for targeted marketing; high-energy digital campaigns should focus on the 25-34 segment for mobile launches, while loyalty programs for appliances might be better aimed at established households in the 35+ range.

Conclusion

Looking at all the data, both Store 2 and Store 11 are performing very well for IslandMart, bringing in nearly identical profits of about MUR 150,000. Even though they make the same amount of money, they do it in different ways.

Store 2 is a tech hotspot. It attracts many young professionals (ages 25–34) who buy mobile phones and accessories. It has a very balanced mix of male and female shoppers. However, the satisfaction scores show some "red flags." While most people are happy, there are a few customers who had very bad experiences. This suggests that while sales are high, the staff might be struggling to keep up with the fast pace, leading to occasional service failures.

Store 11 feels more like a community or family store. It has a much higher number of female shoppers and a more successful delivery service. People here seem more consistently happy, as shown by the higher satisfaction scores and lack of extreme complaints.

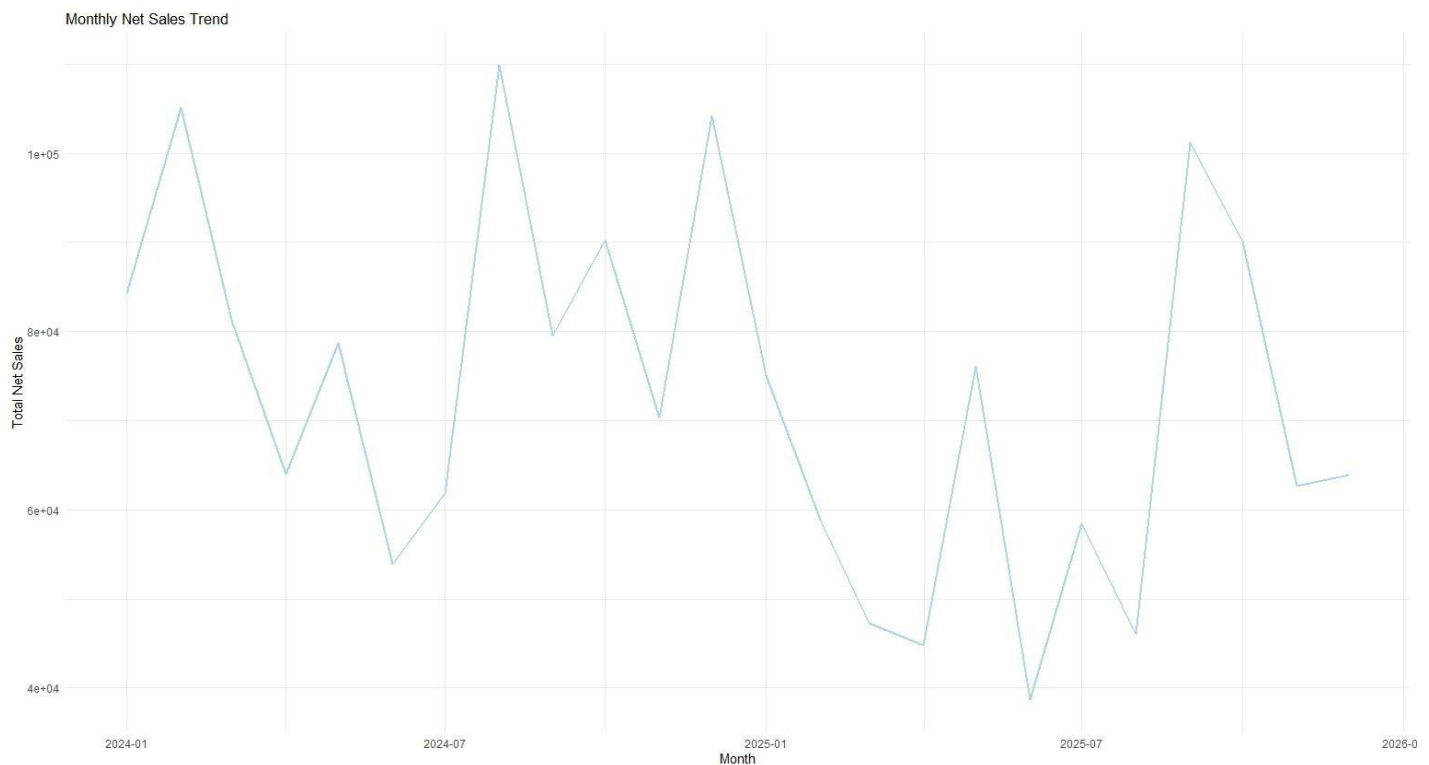
In short, IslandMart has two successful models: a high-energy tech hub and a reliable, service-oriented family store. To grow, management should fix the service gaps at Store 2 and try to grow the delivery business at Store 2 to match the success seen at Store 11.

Task 4 : Time-series + special visualization

Under this section, we will focus more on time-series and spacial visualization of the two MOKA stores.

4.1 Time-series Analysis

The monthly time-series analysis of electronics sales in Moka shows the overall trend in total net sales and total profit. The monthly aggregation reveals observable fluctuations in electronics sales, with potential peak periods that may correspond to festive or promotional seasons. The close alignment between sales and profit suggests stable profit margins across time. This consistent relationship implies that the business maintains effective pricing and cost management strategies throughout the year. However, some fluctuations in both sales and profit can still be observed, which may reflect seasonal purchasing behavior, promotional campaigns, or changes in consumer demand.



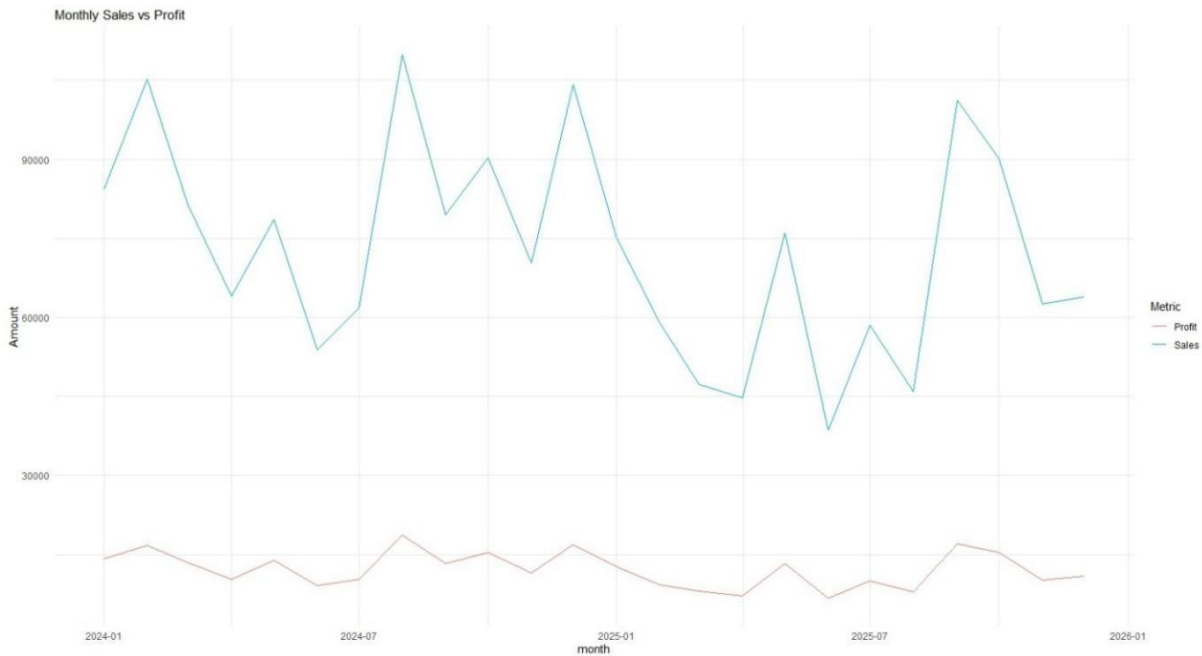


Figure 13: Monthly Sales vs Profit

4.2 Store Performance and Spatial Analysis

The spatial point map visualizes the geographic distribution of store performance, with total sales represented by point size. The larger markers indicate stronger-performing stores, making it easy to identify high-revenue locations. Certain districts show noticeably higher sales levels, which may reflect factors such as greater customer footfall, improved accessibility, or stronger local demand.

Since district boundary polygons were not available, a point map was appropriately selected instead of a choropleth. The spatial distribution reveals clear performance concentration across districts. Black River appears as the dominant district, while Moka ranks third in total gross profit contribution.

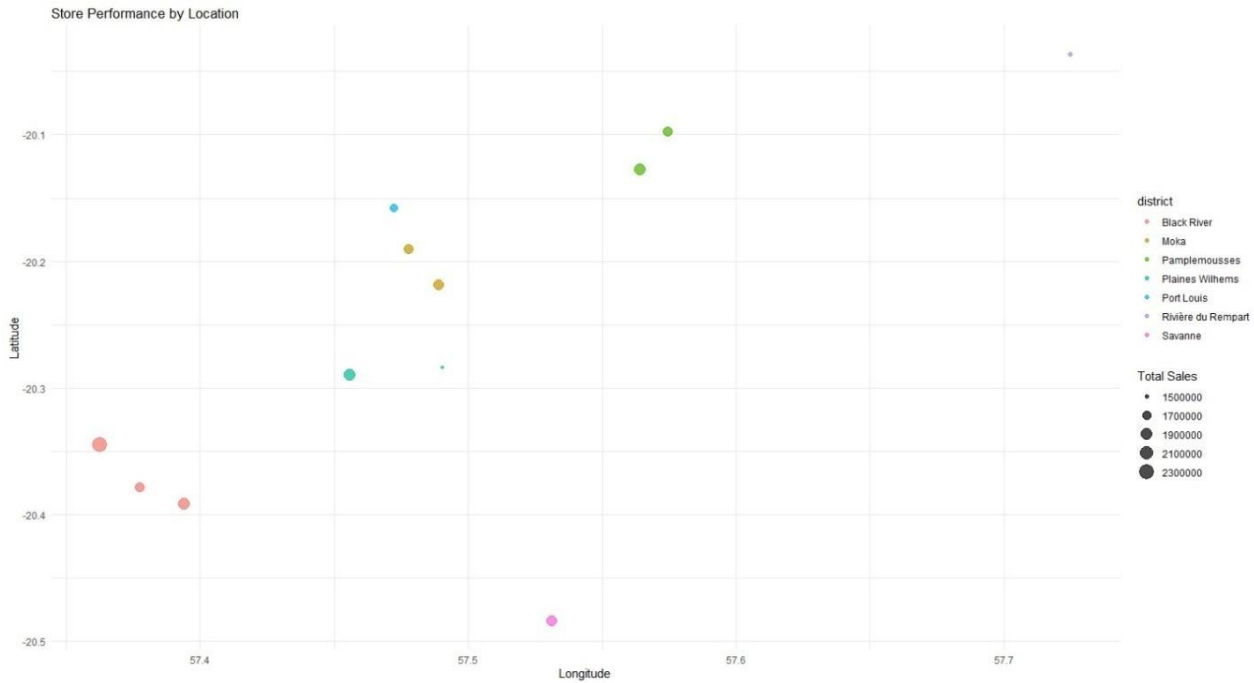


Figure 14: Store performance by Location

Within Moka, the relative marker sizes confirm the time-series findings: Store 11 contributes more substantially to district revenue than Store 02. This spatial clustering of higher-performing stores suggests that geographic and demographic factors play a significant role in revenue generation.

The spatial map also highlights the uneven performance distribution across the retail network. Higher-performing districts may benefit from stronger purchasing power, improved accessibility, or denser commercial activity. By highlighting the Moka district in the below map, it enables a clearer comparison with other districts and helps assess its relative performance within the overall retail network.

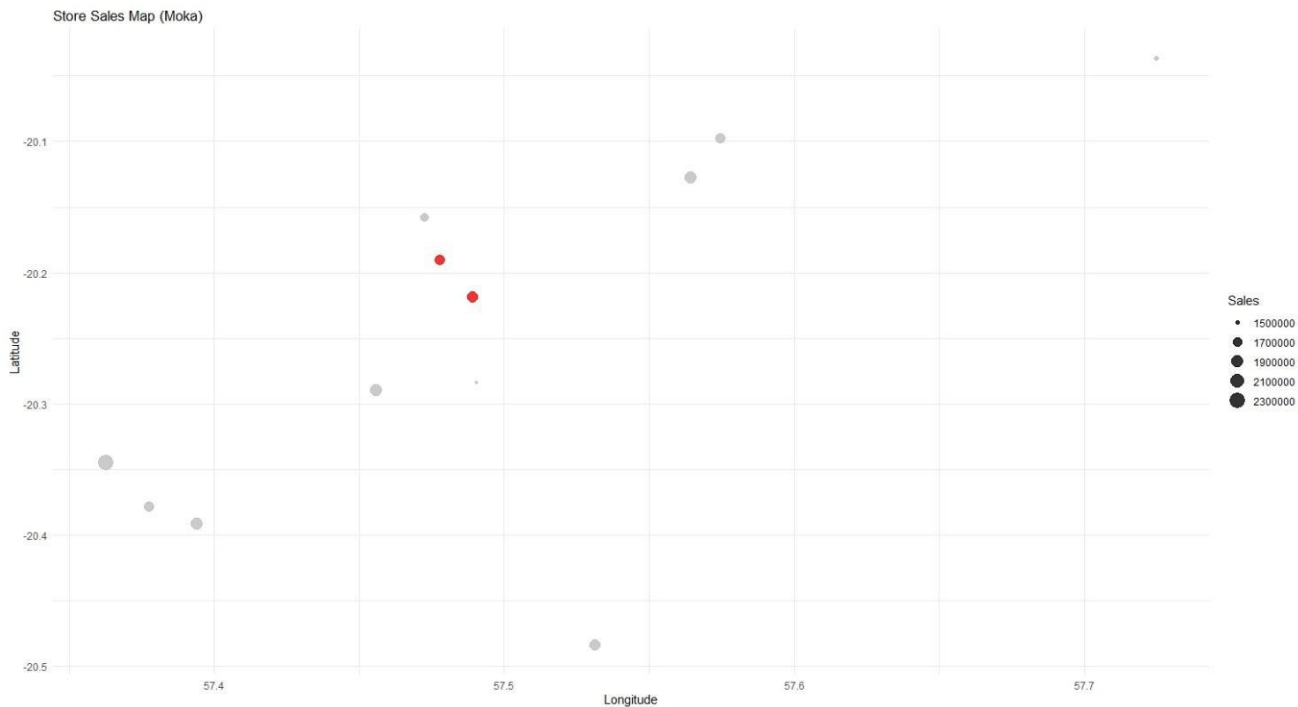


Figure 15: Store Sales of Moka(highlighted)

However, the two maps reflect absolute sales performance rather than efficiency. Sales are not normalized for store size, operating costs, or local population density, meaning higher revenue does not necessarily imply superior operational productivity.

The following limitations must be acknowledged:

1. The monthly aggregation used in Figures 1 to 4 conceals intra-month patterns such as weekend spikes or short promotional surges.
2. The dataset does not extend across multiple years, limiting the ability to confirm recurring seasonal cycles with statistical certainty.
3. The analysis does not incorporate external macroeconomic factors, competitor behavior, or marketing campaign timing, which may influence fluctuations. Regarding Figure 5, spatial comparison based on marker size may visually exaggerate differences in magnitude, and missing coordinate data for certain stores slightly reduces completeness.

Task 5 : Power BI dashboard + storytelling

This part answers the following questions with the use of Power BI dashboard. An overview of the dashboard is included in

Q1: Which district/store drives the highest net sales and gross profit, and how does it change over time?

This report analyses district Moka, focusing on the Electronics product category, and identifies which store drives the highest net sales and gross profit, as well as how these performance indicators evolve over time. The analysis is based on transactional retail data and was visualised using a Power BI dashboard.

Three key performance measures were created using DAX: Total Net Sales, Total Gross Profit, and Profit Margin. The dashboard shows that overall business performance for the Electronics category amounts to Rs 1.75M in total net sales and Rs 292.62K in total gross profit, resulting in a profit margin of 0.17 (17%). This indicates a moderately strong level of profitability.

At the district level, Moka is the highest-performing district in the Electronics category, contributing the full Rs 1.75M in net sales and Rs 292.62K in gross profit. This confirms that Moka is the primary driver of both revenue and profitability within this category.

At the store level, IslandMart Store 11 emerges as the top-performing store for Electronics, generating approximately Rs 874.06K in net sales and Rs 146.58K in gross profit. This store clearly dominates performance, significantly outperforming other stores such as IslandMart Store 02, which contributes a smaller share when selected in the slicer.

The time-series analysis of net sales and gross profit within the Electronics category shows a declining trend over time. This downward pattern suggests a reduction in sales volume or demand for electronic products over the period analysed.

The use of slicers for district, store, sales channel, payment method, and product category enables dynamic analysis. For instance, selecting IslandMart Store 11 highlights its dominant contribution in Electronics, while filtering by different channels or payment methods may reveal additional performance differences.

Overall, the analysis indicates that Moka district and IslandMart Store 11 are the key drivers of business performance. However, the declining trend in both net sales and gross profit over time may be a concern and suggests the need for strategic actions such as promotional campaigns or operational improvements to sustain growth.

Q2: Which categories are most profitable vs most sold (volume), by channel?

The dashboard shows clear differences between product categories when comparing sales value, profit, and volume across channels (Delivery, In-store, Online). In terms of Total Net Sales, Electronics is by far the dominant category, contributing the largest share across all channels, especially in-store. Fashion, Home, and Baby follow at a much lower level, while Grocery, Beauty, and Pharmacy generate relatively small sales values compared to Electronics.

However, when looking at Total Quantity (volume sold), the pattern is completely different. Grocery is the highest by a large margin, with the strongest volume sales, followed by Pharmacy, Home, Baby, and Fashion. Electronics, despite being the top in revenue, ranks much lower in volume sold, showing that it sells fewer but higher-value items. Beauty remains the lowest in both sales and volume.

In terms of profitability (Gross Profit), Electronics again leads significantly, generating the highest profit contribution. Fashion and Home contribute moderate profit levels, while Grocery and Pharmacy show high volume but comparatively low profit. The bubble chart also suggests that In-store channel is the strongest contributor to gross profit, with Online and Delivery trailing behind.

The key insight is a clear disconnect between volume and profitability. Grocery and Pharmacy are the most “sold” categories in terms of quantity, but they do not translate into high profit. On the other hand, Electronics is the most profitable category despite lower volume, meaning it is likely to have higher margins or higher ticket prices.

Channel performance also matters: In-store sales drive most of the profit, especially for high-value categories like Electronics. Online and Delivery channels appear to contribute less to overall profitability, even though they support volume-driven categories like Grocery.

The business should focus on protecting and expanding Electronics in the In-store channel, as this is the strongest profit driver. At the same time, Grocery and Pharmacy should be optimized for efficiency either through cost reduction or upselling strategies to improve margins given their high volume but low profitability.

Additionally, there is an opportunity to strengthen Online and Delivery channels, especially by encouraging higher-margin categories (like Electronics and Fashion) to perform better digitally. This could balance the current over-reliance on in-store profitability and improve overall channel performance.

Q3 : How do channel and payment method relate to basket value, returns, or satisfaction?

The first set of visuals focuses on basket value, and the analysis shows that online transactions tend to generate slightly higher average basket values (Rs 4.18K) compared to in-store (Rs 4.11K) and delivery (Rs 4.09K) channels, indicating that customers shopping online may purchase in larger quantities or higher-value items.

Similarly, when comparing the payment methods, electronic payments such as card and bank transfer are associated with higher basket values, indicating that customers using digital payment options may be more likely to spend more per transaction.

Customer satisfaction is analyzed across channels and in relation to delivery time. The results indicate that satisfaction scores are relatively consistent across all channels, with only minor variation, which suggests that the overall customer experience is stable regardless of how the purchase is made. Furthermore, the scatter plot examining delivery time against satisfaction reveals no strong relationship, indicating that delivery delays may not be the primary driver of dissatisfaction.

Return behavior is also analysed, showing that return rates are slightly higher for online (10.00%) compared to in-store purchases(8.14%). This may be due to customers being unable to physically inspect products before purchase, leading to a higher likelihood of returns. This highlights a potential operational area for improvement, particularly in managing expectations for online orders.

It is important to note that the satisfaction score variable contained “NA” values representing customers who did not participate in the survey. These were treated as structural missing values and converted to null values in Power BI. As a result, average satisfaction calculations exclude non-responses and reflect only valid customer feedback. The three measures were created using DAX: Average Basket Value, Average Satisfaction Score and Return Rate.

Q4: *Where are operational pain points (delivery times) concentrated by district and over time?*

Delivery performance highlights significant operational inefficiencies, with 38.38% of deliveries (around 1,083) arriving late, despite an average delivery time of 55.14 minutes. While this average suggests moderate overall performance, it masks a high level of inconsistency, as nearly two in every five deliveries fail to meet the expected service standard.

These delays are not evenly distributed but are concentrated in specific districts. The clustered column chart shows that Black River, Pamplemousses, and Savanne have the highest late delivery rates (around 39%), making them the primary operational pressure points. In addition, Moka and Plaines Wilhems contribute a substantial number of delays, indicating broader system strain. In contrast, Port Louis and Rivière du Rempart record fewer delays and better performance, positioning them as useful benchmarks for best practices.

This geographical pattern is further supported by the map visual, which shows a clear concentration of delays in high-risk districts through larger and more intense markers. The use of RAG conditional formatting enhances this insight by clearly distinguishing high-risk areas (red) from better-performing districts (green), allowing for quick and effective interpretation.

Over time, the line chart (time trend) reveals that delayed deliveries increase during late 2024 and early 2025, indicating that performance declines during peak demand periods. This suggests that delivery delays are influenced not only by location but also by capacity constraints and rising demand over time.

From an operational perspective, the key issue is performance variability rather than average delivery time. To address this, improvement efforts should focus on reducing delays in high-impact districts such as Black River, Pamplemousses, and Savanne, as these areas contribute most to the total number of late deliveries. At the same time, the observed time trend highlights the need for improved capacity planning and demand forecasting,

particularly during peak periods. Learning from better-performing districts like Port Louis could also help improve overall efficiency.

This analysis assumes that deliveries exceeding 60 minutes are classified as late and that district-level data accurately reflects overall performance. However, it is important to note that aggregated data may hide short-term fluctuations.

Overall, delayed deliveries are both geographically concentrated and time-dependent, requiring a combination of targeted district-level actions and improved system-wide planning to enhance delivery reliability.

Appendix 1

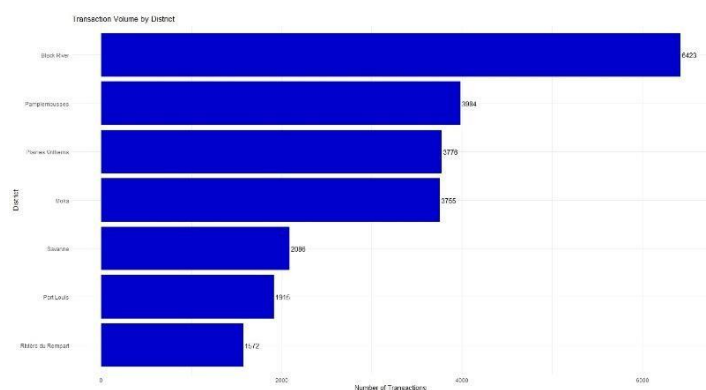
Table 1: Cleaning Log

SN	Variable	Issue	Evidence	Action taken	Impact
1	Transaction_id	Duplication of ids	Duplication of transaction_id values were found during data auditing	Removed duplications	No double counting when will analyse sale and profit
2	discount_pct	Missing values Values outside discount range (0-0.35)	Missing values were identified using colSums(is.na()) Values outside the discount range were found	Replace missing values to 0 (assuming no disco) Removed Values outside range 0-0.35	Ensuring accuracy in net sales calculations It will help in providing realistic pricing analysis
3	quantity, unit price MUR	Included invalid values	Records where quantity ≤ 0 or unit_price_mur ≤ 0	Removed invalid rows	Remove the possibility of invalid transactions
4	gross_sales_mur, net_sales_mur, cost_mur, gross_profit_mur	Financial subtle inconsistencies	net sales differ from gross_sales*(2-discount)	Adress those inconsistencies by using the formula: net sales = gross*(1-discount)	Removed inconsistencies and standardized for financial analysis
5	delivery times	Missing for in store	16626 values were missing, in store channel	Replaced missing values as NA	Delivery time relates mainly to delivery and online. In-store should not have delivery time
6	net sales mur	Outliers present	Top 1 % of net_sales was reflecting outliers	Removed the extra 1 % outlier to avoid distortion	Data more consistent for financial reporting and for visual representations
7	Date/datetime	Inaccuracy data format- Data structure as character	Date and datetime were classed as character	Removed invalid dates Convert date /datetime to correct format	More consistent data Change class for datetime for time series analysis
8	satisfaction score	Values missingness	Around 10% Missing mostly around all channels	No change was made. Left it as NA only	Left as it is as not all customers are surveyed.
9	store_lat, store_lon	Store coordination missing	140 rows from colSums(is.na())	Excluded from special map only	Does not affect any financial analysis(if any)

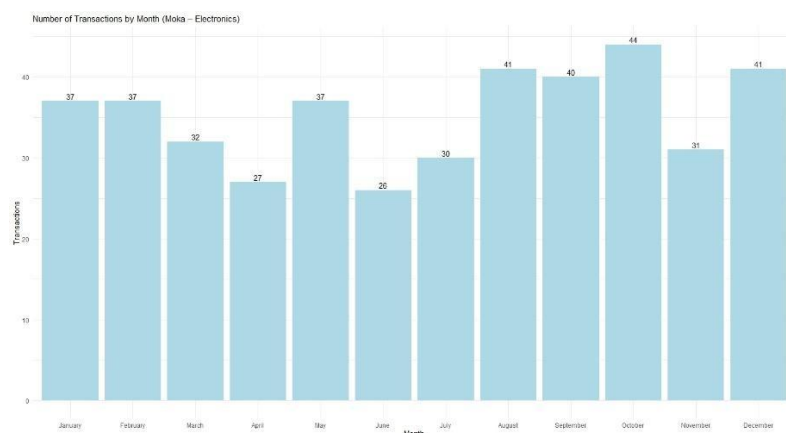
Overall Data Summary – District, No. of Stores and No. of transactions

District	No. of Stores	No. of transactions
Black River	3	6423
Pamplemousse	2	3984
Plaine Wilhems	2	3776
Moka	2	3755
Savanne	1	2086
Port Louis	1	1915
Riviere du Rempart	1	1572

Transaction count by Districts



Moka Monthly transactions count



Summary Focus Data Missingness : MOKA/Electronics

Table 2: Moka Missingness values

delivery_minutes	satisfaction_score	transaction_id	datetime
258	45	0	0
date	year	month	year_month
0	0	0	0
year_quarter	store_id	store_name	district
0	0	0	0
store_lat	store_lon	customer_id	customer_gender
0	0	0	0
age_band	customer_segment	channel	payment_method
0	0	0	0
Campaign	product_category	product_subcategory	quantity
0	0	0	0
unit_price_mur	discount_pct	gross_margin_pct	gross_sales_mur
0	0	0	0
net_sales_mur	cost_mur	gross_profit_mur	returned_flag
0	0	0	0
calc_net_sales	diff		
0	0		

Task 2: Summary Statistics Table

Table 3: Summary Statistics

Variable	Mean	Median	Min	Max	Skewness	Kurtosis
Net Sales (MUR)	4128.25	4015.39	1627.71	6902.32	0.25	2.29
Gross Profit (MUR)	691.76	657.98	229.70	1627.53	0.57	3.05
Delivery Minutes	31.90	30.00	5.00	79.00	0.68	3.08
Satisfaction Score	4.04	4.00	2.70	5.00	-0.06	2.54

Grouped Distributions (Violin Plots)

From the violin plots, the following conclusions have been derived:

1. Net Sales are broadly similar across all segments, with Returning customers showing a slightly higher concentration of peak sales around 7,000 MUR
2. Gross Profit is most consistent for VIP customers, while Returning customers exhibit the highest potential upside with a significant "long tail" of high-profit transactions.
3. Delivery times vary by channel: Online orders are generally faster and concentrated around 25 minutes, Delivery orders are longer and more variable, while the In-store channel has no recorded delivery minutes, as these values are missing from the dataset.

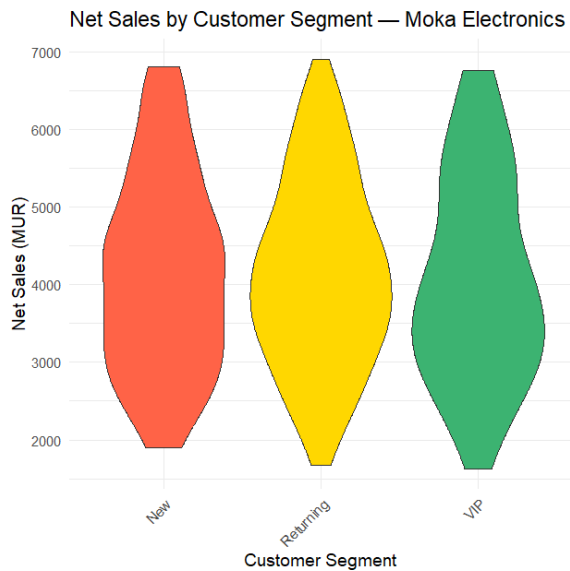


Figure 1c: Net Sales by Customer

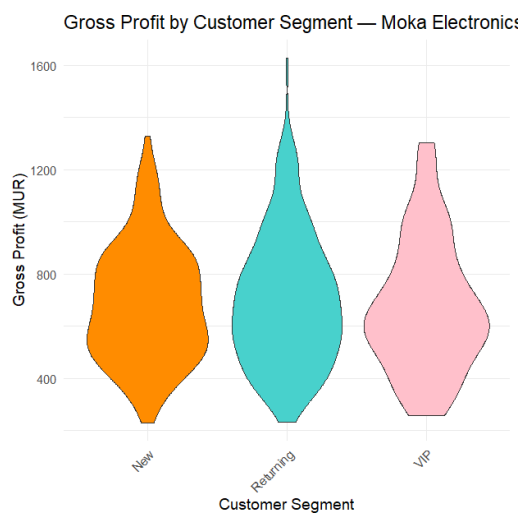


Figure 17: Gross Profit by Customer Segment

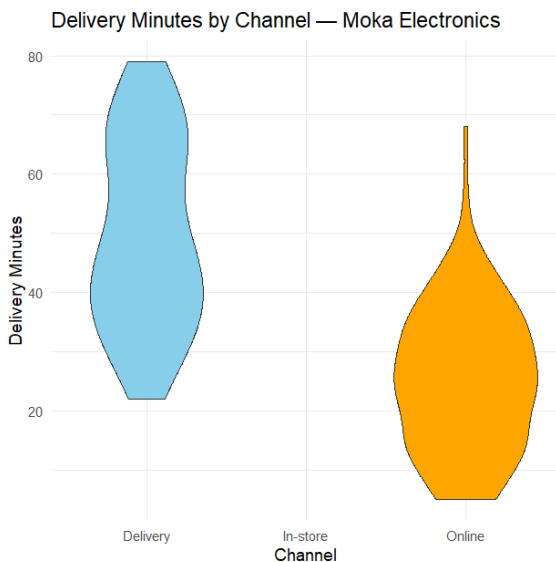


Figure 18: Delivery Minutes by Channel

Overview of PowerBI Dashboard – MOKA District with Product Category Electronics

